



Assessing grassland cultural ecosystem services supply and demand for promoting the sustainable realization of grassland cultural values

Lan Wang^{a,b}, Lin Huang^{a,*}, Wei Cao^a, Jun Zhai^c, Jiangwen Fan^a

^a Key Laboratory of Land Surface Pattern and Simulation, Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences, Beijing 100101, China

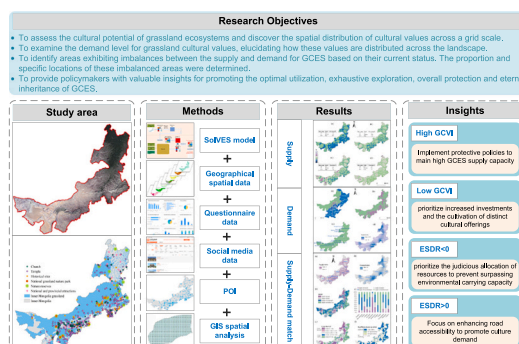
^b University of Chinese Academy of Sciences, Beijing 100049, China

^c Satellite Application Center for Ecology and Environment, Ministry of Ecology and Environment of the People's Republic of China, Beijing 100094, China

HIGHLIGHTS

- GCES were quantified at a grid scale by integrating model and multi-source data.
- Ecotourism with 46.0 % high-value areas was predominant way to realize GCES value.
- GCES exhibited a significant supply-demand imbalance with 65.99 % oversupplied.
- Clustered zones were generated to facilitate the implementation of targeted measures.

GRAPHICAL ABSTRACT



ARTICLE INFO

Editor: Paulo Pereira

Keywords:

Grassland cultural ecosystem services
Supply and demand relationship
SolVES model
Social media
Inner Mongolia Autonomous Region

ABSTRACT

Grasslands deliver essential provisioning and regulatory ecosystem services, concomitantly with indispensable cultural services that merit profound consideration. However, grassland cultural ecosystem services (GCES) face a conspicuous knowledge lacuna due to the lack of a unified research framework and quantitative methodology. This study endeavors to fill this gap by quantifying the potential supply and actual demand of GCES, concurrently scrutinizing spatial congruencies and disparities between GCES supply and demand in Inner Mongolia Autonomous Region (IMAR), China. To achieve this, we integrated social survey data, Point of Interest (POI) data, social media data, the Social Values for Ecosystem Services (SolVES) model and GIS Getis-Ord Gi* statistical analysis. Our analysis unveiled grid-scale spatial patterns of GCES supply and demand, furnishing a nuanced high-low ranking of GCES. It transpired that scenic travel holds the highest potential supply of GCES with a high-value area proportion of 46.0 %, while grassland recuperation ranks the lowest. Notably, road accessibility emerged as the most crucial factor influencing GCES patterns. Furthermore, we observed a substantial misalignment in the GCES supply-demand relationship, with 65.99 % of IMAR experiencing excess supply compared to demand and only 20.66 % achieving equilibrium. At a 95 % significance level, hot spots (excess supply) and cold spots (excess demand) accounted for 26.03 % and 22 %, respectively. We propose targeted suggestions that regions with oversupply of GCES should channel efforts toward augmenting road accessibility,

* Corresponding author.

E-mail address: huanglin@igsnr.ac.cn (L. Huang).

<https://doi.org/10.1016/j.scitotenv.2023.169255>

Received 1 July 2023; Received in revised form 6 December 2023; Accepted 7 December 2023

Available online 12 December 2023

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whereas regions grappling with excess demand should prioritize the judicious allocation of resources to avert surpassing the environmental carrying capacity. Our study furnished insights for decision-makers to formulate sustainable development plans pertaining to grassland culture.

1. Introduction

Cultural ecosystem services (CES) have gained prominence as a focal point of study, garnering heightened significance within the realm of ecosystem services research and policy formulation (Plieninger et al., 2015; Bagstad et al., 2017; Yiyao and Kiichiro, 2023). They are defined as the multiple intangible values and non-material benefits provided by ecosystems through experiences, recreation and tourism, which enhance human well-being by meeting physiological, psychological, and emotional needs (Hernandez-Morcillo et al., 2013; Díaz et al., 2019). Different classification frameworks for CES (Supplementary Table S1) have been proposed that, while varying, generally encompass aspects such as aesthetics, inspiration, spirituality and religion, education, recreation and ecotourism (Costanza et al., 1997; Daily, 1997; De Groot et al., 2002; MEA (Millennium Ecosystem Assessment), 2005; Fletcher et al., 2014). CES, as a link between society and nature, often promote comprehensive public understanding and protection of ecosystems, facilitate policy makers to consider their potential cultural added value, avoid bias in ecosystem management processes, and ultimately promote optimal ecosystem management to ensure sustainable socioeconomic development (Daniel et al., 2012; Díaz et al., 2019; Kosanic and Petzold, 2020). Therefore, in order to achieve sustainability, it is critical for decision makers to consider the spatial characteristics of CES to counteract the degradation of ecosystems and to improve human wellbeing worldwide (Burkhard et al., 2012; Díaz et al., 2019; Teoh et al., 2019; Fsa et al., 2021).

However, compared to tangible ecosystem provisioning and regulating services, CES research is challenged by its intangibility and subjectivity, and is difficult to be integrated into ecosystem services research frameworks due to a lack of systematic research on CES and insufficient subjective knowledge of CES (Satz et al., 2013). Additionally, constraints in data availability and methodological weaknesses make its quantification difficult (Plieninger et al., 2015; Shaw et al., 2016; Tenerelli et al., 2016; Small et al., 2017; Teoh et al., 2019). Relevant studies are relatively scarce and CES are often ignored or replaced by a certain type for the whole CES. The mechanism and estimation methods of CES still require further research and improvement. CES have obvious spatial and temporal heterogeneity of supply and demand, and the spatial transfer process, flow direction, flow rate, and flow volume of supply and demand are still unclear (Burkhard et al., 2012; Syrbe and Grunewald, 2017; Wei et al., 2017). Demand evaluation methods are still relatively limited, so their spatial evaluation and mapping research are progressing slowly. Overall, CES face a knowledge gap so multi-scale and cross-scale CES research are desperately needed (Gould and Lincoln, 2017).

In recent years, non-monetary value assessment, spatial mapping and supply-demand research of CES have gained increasing prominence (Bagstad et al., 2017; Karasov et al., 2022; Mitra et al., 2023). Quantitative evaluation of CES currently relies on a range of methodologies, including biophysical modeling (Paracchini et al., 2014; Mitra et al., 2023), SolVES model (Sherrouse et al., 2011; Bagstad et al., 2017; Shiting et al., 2020; Sitong et al., 2023), maximum entropy (Maxent) model (Shan et al., 2019; Zuo and Zhang, 2023), Public Participatory Geographic Information Systems (PPGIS) (De Valck et al., 2016; Crouzat et al., 2022), qualitative comparative analysis (QCA) (Xin et al., 2022), and deep learning (Liao et al., 2023). Additionally, multivariate quantitative data encompass social media data (Gosal et al., 2019; Lingua et al., 2023; Qiqi et al., n.d.), POI data (Shan et al., 2019), location data (Yiyao and Kiichiro, 2023), and remote sensing data (Karasov et al., 2022). The SolVES model, frequently relying on PPGIS results to map

CES scales, examines human attitudes and perceptions regarding ecosystem services for human well-being using a ranking methodology (Bagstad et al., 2017). Social media data serve as a proxy for evaluating interactions between people and nature, particularly in identifying outdoor activities and landscape appreciation (Gosal et al., 2019). They provide evidence of CES utilization in regions with inadequate, disorganized, or irregular statistical data (Toivonen et al., 2019).

CES studies have been conducted in various ecosystem types such as forests, wetlands, agriculture, grasslands and deserts, with a predominant focus on urban green spaces, agricultural lakes, nature reserves and national parks (Sherrouse et al., 2011; Liu et al., 2020; Crouzat et al., 2022; Das et al., 2023). However, limited attention has been given to grasslands, despite them being the most extensive terrestrial ecosystems globally, comprising approximately 24 % of the Earth's total land area, and Chinese grasslands alone accounting for about 35 % of the national land area (Li and Fang, 2017). In the specific grassland region, an enduring nomadic culture has thrived for millennia, shaped by the intrinsic interplay of livelihoods, dialects, historical narratives, and emotional connections intertwined with the bountiful sustenance and natural surroundings of water and grass (Liu, 2010). GCES can be regarded as its cultural genes and spiritual qualities formed during long-term historical development process, including different ecological and cultural landscapes, particular production methods, local cultural language and traditional customs, and other various cultural elements. Moreover, a systematic review of CES indicates that existing research on CES covers mainly Europe and North America. Gaps in this field are evident especially regarding Africa, central and east Asia and Latin America. A major research trend in the field is to expand the geographic scope of research (Kosanic and Petzold, 2020). Therefore, it is imperative to conduct CES research within the grasslands of China.

In summary, with the backdrop of rich cultural potential in grassland regions and a continued scarcity of related research, the imperative for conducting GCES research becomes evident. Given the nascent stage of CES research and its evolving technical methodologies, we propose the following GCES hypotheses: (1) Public preferences, particularly those of non-expert survey participants, mirror potential supply of CES, which can be regarded as the hypothetical maximum yield of a particular area in providing specific cultural services (Burkhard et al., 2012). (2) Visit data obtained from social media sites can represent actual CES demand, which refers to the cumulative consumption or utilization of all cultural services within a specific area over a defined time period (Burkhard et al., 2012). (3) A notable disparity exists between potential supply and actual CES usage, highlighting an imbalance in CES supply and demand. Therefore, our study integrates SolVES model, questionnaire and POI data to model GCES potential supply, utilizes social media data to model the actual use of GCES, and employs geospatial statistical analyses to explore the spatial dynamics of GCES supply and demand. Our research pursues three specific objectives: (1) To comprehensively assess the cultural potential of grassland ecosystems and discover the spatial distribution of GCES supply across a grid scale. (2) To examine the spatial variability in the demand for GCES, elucidating how these values are distributed across the landscape. (3) To identify areas manifesting supply-demand imbalances in GCES, with a determination of the proportions and specific locations of these imbalances.

2. Materials and methods

2.1. Study area

IMAR, situated in Northern China, spans an expansive 1.183×10^6

km², constituting 12.3 % of terrestrial China. Primarily characterized by a plateau-type landform, the region boasts an average altitude exceeding 1000 m. Its temperate continental monsoon climate manifests as semi-humid in the east and semi-arid in the west, with an annual temperature range of 0–8 °C and precipitation varying between 100 and 500 mm. As an essential ecological security barrier and an important national production base for agricultural and livestock products, the natural grasslands encapsulate 0.88×10^6 km². This accounts for 22 % of the nation's total grassland area and 74 % of the region's surface. The 476 grassland types of IMAR representing about 44 % of the total grassland types in China (Li and Fang, 2017). Progressing from east to west, the sequence includes meadow grasslands, typical grasslands, desert grasslands, and steppe desert (Niu, 2000). The eight grasslands of IMAR comprise Alashan, Wulate, Eerduosi, Wulanchabu, Xilinguole, Zhaowuda, Keerqin, and Hulunbeier. A diversity of tourism resources in IMAR can be revealed in Fig. 1.

2.2. GCES connotation

We have summarized various subtypes of grassland culture to understand GCES connotation (Supplementary Table S2). Drawing on prior research on CES categorization (Supplementary Table S1) and acknowledging potential interconnections and overlaps in the values they represent, this study classified the GCES into four overarching categories: scenic travel value (ST-V), spiritual belief value (SB-V), science-education-history value (SEH-V), and grassland recuperation value (GH-V). ST-V encompasses tourism experiences and recreational enjoyment derived from a region's natural ecology and cultural elements through tourism industry development. SB-V fulfills spiritual needs via religious practices, fostering personal growth, exemplified by Wusutu Temple and Zhaohua Temple. SEH-V integrates local customs, folklore, and folk culture into literary and artistic works while supplying creative material for popularizing traditional nomadic culture and supporting

nomadic cultural experiences, typified by the Yuan Shangdu Site. GH-V provides leisure, connecting visitors with nature, relieving stress, enhancing health, and offering cultural enrichment. It deepens appreciation for nomadic culture and provides a holistic wellness experience for those seeking respite from the fast-paced world.

2.3. Data collection

In our study, we employed a diverse datasets comprising questionnaire responses, geospatial environmental information, POI data, and social media data, with details about data formats and sources provided in Table 1. Survey data for the SolVES model was gathered via Wenjuanxing (<https://www.wjx.cn/>). Access to the questionnaire design and content can be found in Supplementary Questionnaire S1 and S2, respectively. Out of 320 questionnaires obtained online, 272 met the established criteria, resulting in an 85 % effective response rate. We utilized python for web scraping social media data from prominent platforms, such as Dianping and Mafengwo, to assess GCES demand. Furthermore, we enhanced our understanding of demand by sourcing primary information on domestic tourists originating from regions outside IMAR from the Inner Mongolia Cultural and Tourism Department's website (<https://wlt.nmg.gov.cn/>). Finally, we chose the following factors, including Distance to Roads (DTR), Distance to Water (DTW), Distance to Grassland (DTG), Elevation (ELEV), Slope, Land Use types (LANDUSE), and Landforms (LANDFORM), as environmental input elements for SolVES model (Supplementary Fig. S1). To ensure that the spatial range and projection of each environmental raster data were consistent with the study area in shapefile format, we conducted mask extraction using the boundary data of the study area. To enable a comprehensive investigation into the interplay between environmental factors and social values, we proceeded to standardize the resolution, as well as the number of rows and columns, for each environmental raster dataset. The resolution was uniformly set to 450 m, and the number of

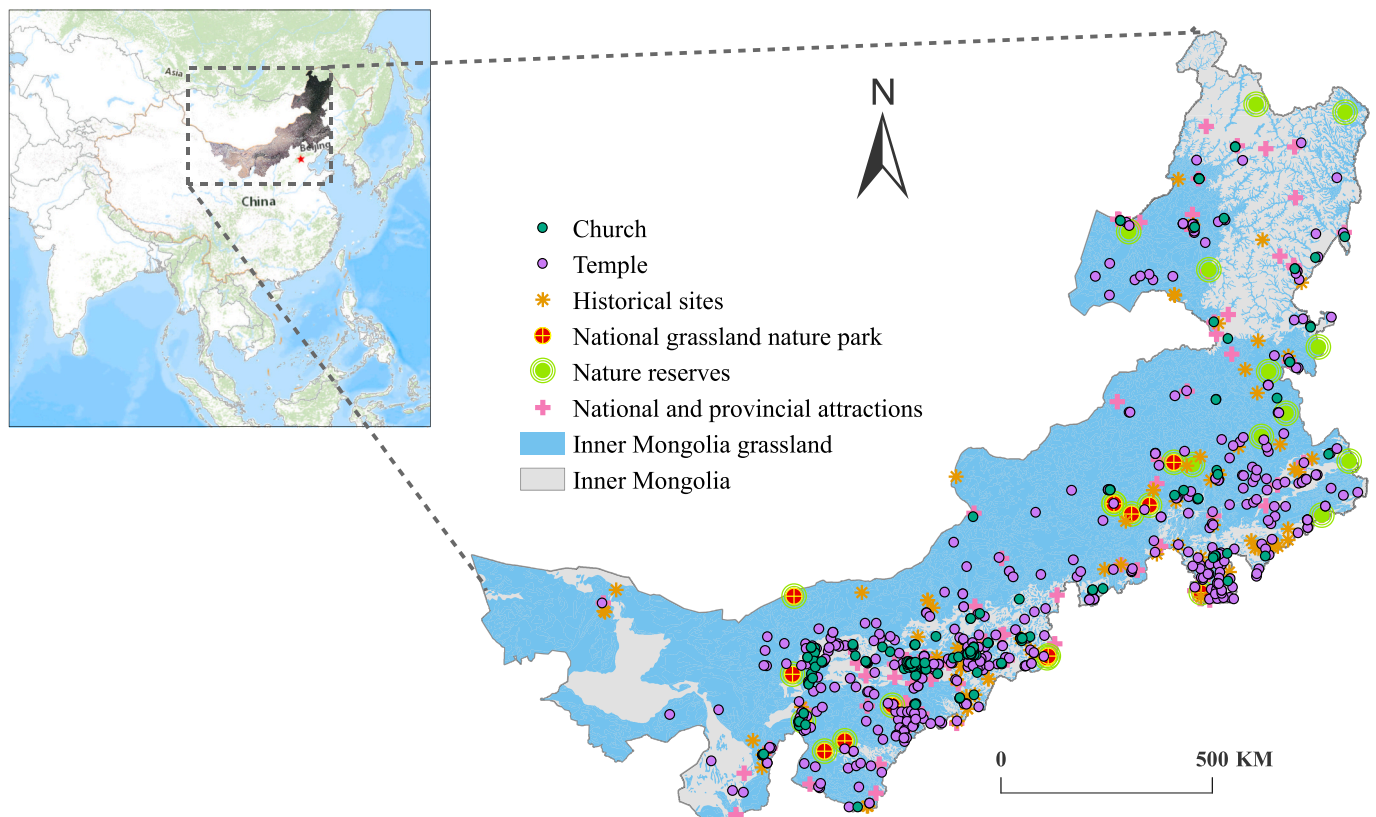


Fig. 1. Location of the study area and distribution of major POI.

Table 1
Main data used in this study.

Name	Format	Description	Resolution	Source
ELEV	Tiff	Digital Elevation Model (DEM) in meters	250 m	Geospatial Data Cloud Platform (https://www.gscloud.cn)
SLOPE		slope	250 m	Slope calculation based on DEM
LANDUSE		Land Use types	30 m	Resource and Environment Science Data Center of the Chinese Academy of Sciences (https://www.gscloud.cn)
LANDFORM		Land surface form data	1000 m	Resource and Environment Science Data Center of the Chinese Academy of Sciences (https://www.gscloud.cn)
DTR		Distance to roads (horizontal distance to nearest road in meters)	250 m	Euclidean distance calculation based on road network data (National Basic Geographic Information Center website (http://www.ngcc.cn/ngcc/))
DTW		Distance to water (horizontal distance to nearest water body in meters)	250 m	Euclidean distance calculation based on drainage data (Resource and Environment Science Data Center of the Chinese Academy of Sciences (https://www.gscloud.cn))
DTG		Distance to grassland (horizontal distance to nearest grassland in meters)	250 m	Euclidean distance calculation based on grassland data (Resource and Environment Science Data Center of the Chinese Academy of Sciences (https://www.gscloud.cn))
VALUE ALLOCATION	Table	Assuming numerical values (total of 100) are allocated to each value type	–	Value assigned from questionnaire
SURVEY_POINTS	ShapeFile	Social value point layer	–	Digitization of value points from questionnaires
Social media data	ShapeFile	Tourists information from social media data	–	Dianping (https://www.dianping.com/) and Mafengwo (https://www.mafengwo.cn/)
POI data	ShapeFile	Major POI in the study area	–	Gaode Map API (https://lbs.amap.com/)

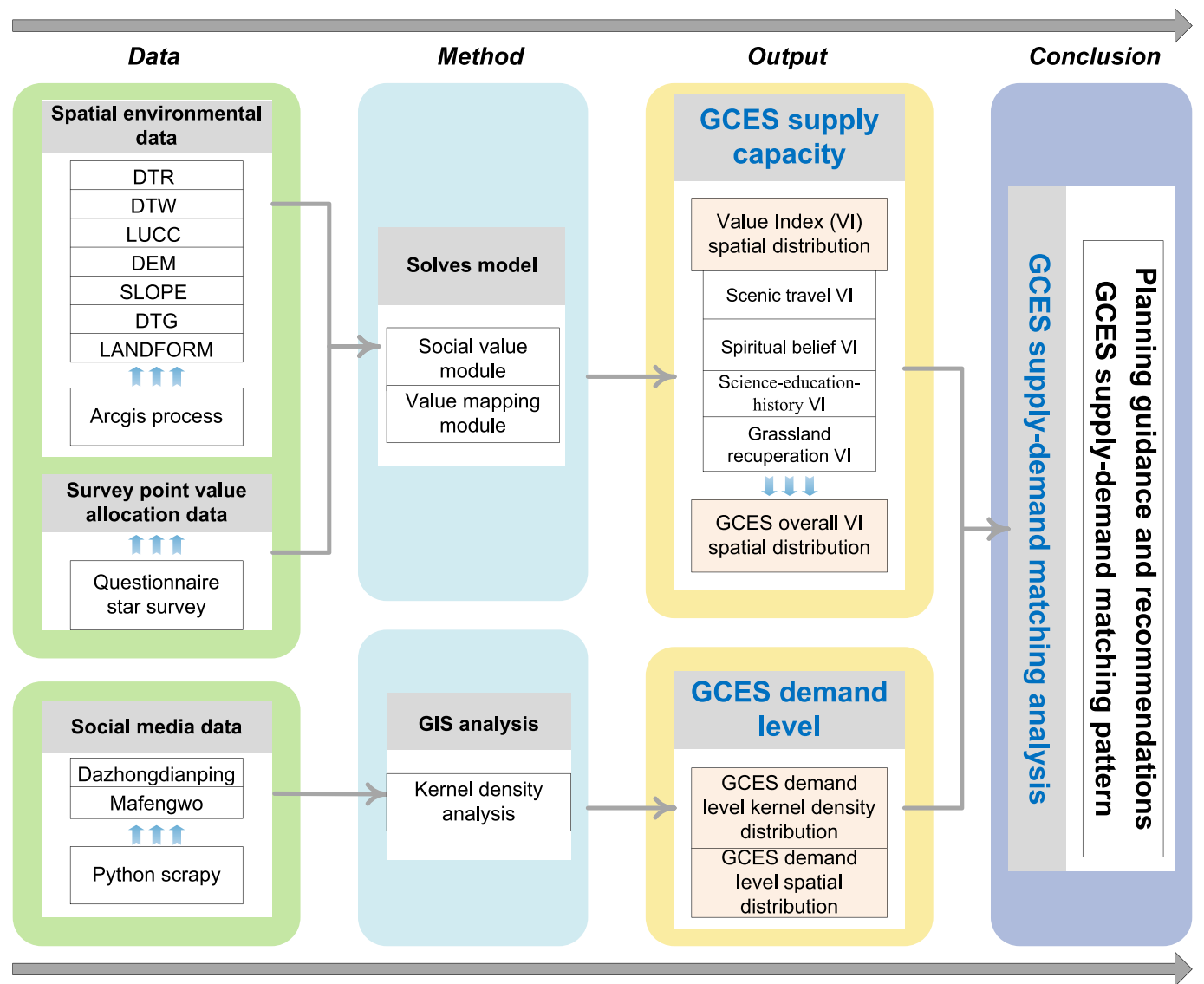


Fig. 2. Framework for grassland cultural ecosystem services (GCES) study.

rows and columns were standardized to 4712 and 4162, respectively.

2.4. Methods for GCES quantification

This study focused on two primary dimensions: the potential supply of GCES, assessed using the SolVES model in conjunction with questionnaire data and geographical spatial information; and the actual demand for GCES, gauged through the analysis of social media data. Ultimately, a GIS-based spatial analysis was employed to evaluate the concordance and disparities between GCES supply and demand. The framework of this study was presented as shown in Fig. 2.

2.4.1. Quantifying GCES supply based on SolVES model

This study used the SolVES model to map the value of GCES. It was developed by Colorado State University and the US Geological Survey for assessing ecosystem cultural services, and has been successfully applied to evaluate CES (Sherrouse et al., 2011). The model consists of three modules: social value, value mapping, and value transfer mapping. The model necessitates input data comprising environmental data layers, survey data, and delineation of the research area boundary. Environmental data, tailored to the actual conditions of the research area, typically includes variables such as elevation, landform, land-use types, and distance to roads. Survey data encompasses information on value distribution and vectorization of value points.

The CES quantification mainly relies on the functions of the social value module and the value mapping module (Fig. 3). The social value module functions by overlapping the valued weight of each cultural service point surveyed. Subsequently, kernel density calculations are applied to these surveyed points, generating kernel density surfaces for each type of cultural service point. Comparisons are made within and between each type of kernel density surface to determine the maximum density value of each type of surface. The value mapping module uses the kernel density surface obtained from the previous module, normalized based on the maximum density value, to generate layers of each type of cultural service within the range of 0–10, with the maximum value of each type of cultural service layer being the maximum value index of that type. Then, SolVES transfers the survey data and environmental data to the built-in Maxent model, which generates a logic output layer for each type of cultural service based on these input data. After multiplying this layer with the maximum value index of each type of cultural service, the final distribution map of the value index of each

type of cultural service can be obtained. Subsequently, the spatial distribution maps of the value index of four types, including ST-V, SB-V, SEH-V, and GH-V, are obtained separately. The mean value index for these four types is calculated to obtain the total GCES supply.

The value mapping function of the SolVES model is mainly provided by the Maxent model. SolVES uses Maxent to generate a model that describes the relationship between environmental variables and the occurrence of given observation points (Phillips and Dudík, 2008). The Maxent model can calculate the area under the ROC curve of the SolVES model (AUC value) to evaluate the suitability of the SolVES model and to verify the reliability of the evaluation results. If the AUC is 0.5, the model is random. The higher the AUC value, the better the model performance (0.5–0.7, 0.7–0.9, and > 0.9 represent low, medium, and high accuracy, respectively) (Swets, 1988). In addition, Maxent can output the influence of variables in the prediction results, including the relative contribution and importance of each variable and the response curve.

2.4.2. Mapping GCES demand by social media data

CES demand refers to people's willingness, preferences and needs to use cultural services (Schrter et al., 2014). Ongoing research is utilizing social media data to assess aesthetic and entertainment values, with potential application in quantifying demand for CES (Yoshimura and Hiura, 2017). Since CES demand can be viewed as actual usage, we quantified it by tracking the number of visitors to specific scenic locations. Data sourced from websites underwent rigorous processing, including filtering out empty spatial information, merging datasets, and geographic encoding. The measured GCES demand was normalized into the range of 0–10 to ensure consistency with GCES supply, and the specific calculation formula is:

$$N_p = N_{pd} + N_{pm} \quad (1)$$

$$D_p = 10 * (N_p - N_{pmin}) / (N_{pmax} - N_{pmin}) \quad (2)$$

where N_p is the number of people going to location p , N_{pd} and N_{pm} are the number of people going to a location obtained through the Dianping website and Mafengwo website respectively, D_p is the demand index for location p , and N_{pmin} and N_{pmax} are the minimum and maximum values in the number of people.

To pinpoint GCES demand hotspots, we conducted a kernel density analysis, transforming point elements into raster data. This effective method calculates the density of spatial elements within their

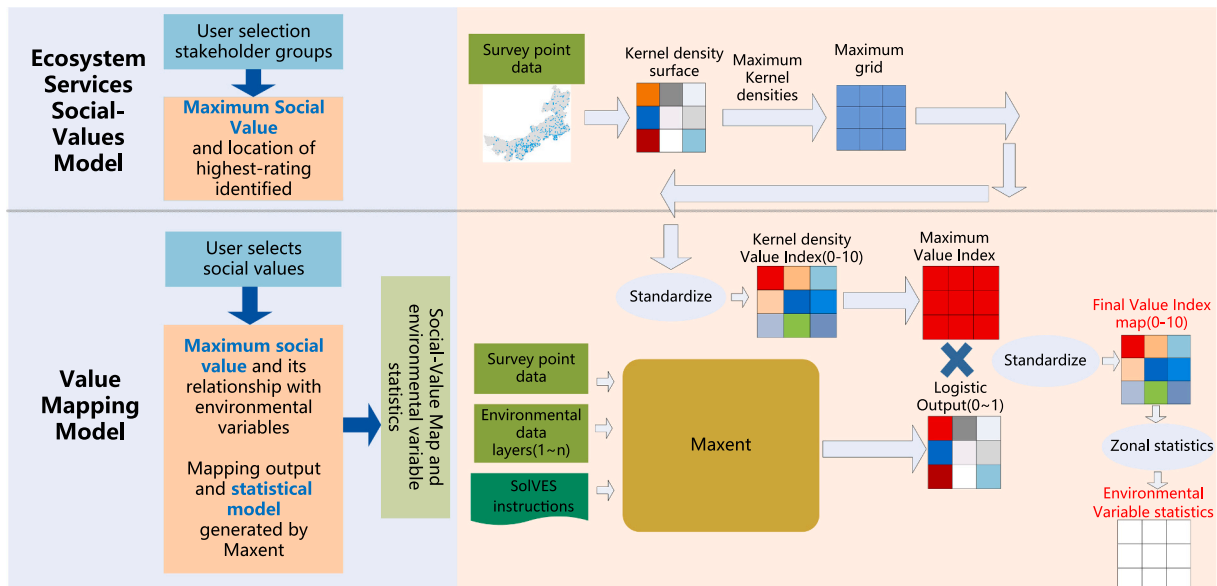


Fig. 3. Function diagram of the Social Values for Ecosystem Services (SolVES) model.

surroundings, revealing continuous distribution patterns through assigned kernel density values. Leveraging the ArcGIS 10.8 tool for the analysis, we derived the spatial distribution density of grassland demand points, enabling the precise identification of areas with the highest concentration of GCES demand.

2.4.3. GCES supply-demand ratios

We identified spatial matches and mismatches of GCES in the territory by calculating the Ecological Supply-Demand Ratio indicator (ESDR), which was established by Li et al. (2016), indicating a surplus, balance, or deficit (Chen et al., 2019; Lorilla et al., 2019; Liu et al., 2020; Jiang et al., 2021; Pan et al., 2021). The ESDR is calculated as follows:

$$ESDR = \frac{ESS - ESD}{(ESS_{max} + ESD_{max})/2} \quad (3)$$

where ESS and ESD are the supply value index and the demand rank index for GCES; ESS_{max} and ESD_{max} represent the maximum value of supply and demand, and are extracted from the corresponding ESS and ESD spatial layers, respectively. A positive ESDR value indicates an GCES supply surplus, a negative value indicates that demand exceeds supply, and a value of zero indicates supply-demand balance.

2.4.4. Spatial clustering of GCES supply-demand mismatch

While ESDR can provide an intuitive visualization of spatial relationship between GCES supply and demand at a grid scale, there exists a

phenomenon of fragmentation. To examine this matching on a broader scale, we introduced the Getis-Ord G_i^* statistic (Getis and Ord, 1992) to perform a spatial supply-demand matching pattern clustering analysis based on the ESDR values, which identifies statistically significant high and low clusters of supply-demand metrics (Liu et al., 2020). Getis-Ord G_i^* is calculated as Eq. (4):

$$G_i^* = \frac{\sum_{j=1}^n w_{ij} x_j - \left[\frac{\sum_{j=1}^n x_j}{n} \right] \sum_{j=1}^n w_{ij}}{S \sqrt{\frac{n \sum_{j=1}^n w_{ij}^2 - \left(\sum_{j=1}^n w_{ij} \right)^2}{n-1}}} \quad (4)$$

$$S = \sqrt{\frac{\sum_{j=1}^n x_j^2}{n} - \left[\frac{\sum_{j=1}^n x_j}{n} \right]^2} \quad (5)$$

where n is the number of spatial features; w_{ij} is the distance between features i and j ; x_j is the value of ESDR; and S is calculated as Eq. (5).

The Getis-Ord G_i^* analysis yields a Z value, where $Z > 1.96$ signifies a hotspot, indicating a cluster of high values representing areas of significant supply exceeding demand. Conversely, $Z < -1.96$ suggests a cold spot, representing areas with considerably higher demand than supply. When $-1.96 < Z < 1.96$, no significant clustering pattern is observed. We categorize the Z -scores into five groups: $Z < -1.96$ (cold spot), $-1.96 < Z < -1.65$, $-1.65 < Z < +1.65$, $+1.65 < Z < +1.95$ and $Z > 1.96$ (hot spot), assigned to zones 1–5 respectively. Zone 1–5

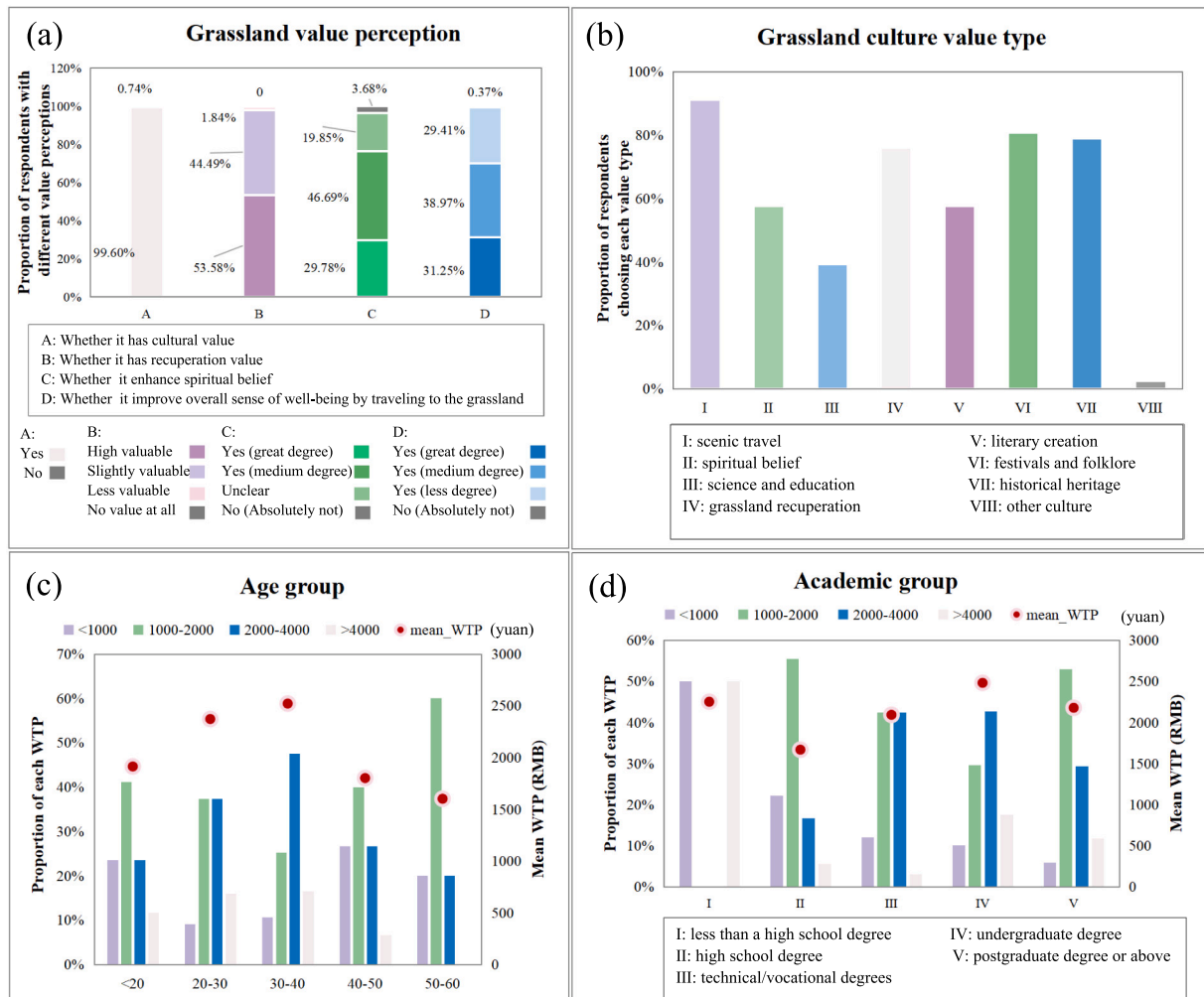


Fig. 4. Respondents' basic cognition for grassland culture. (a) Proportion of respondents with different value perceptions, (b) Proportion of respondents choosing different value types, (c) Proportion of each willingness to pay (WTP) for different age groups, (d) Proportion of each WTP for different academic groups.

demonstrates a gradient from high to low demand values and from low to significantly higher supply values.

3. Results

3.1. Perceived notions for GCES among participants

A robust 99.6 % of respondents acknowledged the cultural significance of grasslands, encapsulating scenic travel, festivals, folklore, historical heritage, grassland recuperation, spiritual belief, literary inspiration (Fig. 4). Among these, scenic travel value was the most frequently selected, contrasting with the diminished preferences observed for science and education. Over half of participants viewed grasslands as highly conducive to recuperation, while a staggering 98 % deeming them to be either very or relatively valuable for supporting health. Moreover, 76.47 % of respondents espoused the belief that grasslands have the potential to augment spiritual well-being. This collective sentiment extended to 99.63 % of respondents, expressing the conviction that a visit to the grasslands could significantly elevate their overall sense of well-being, with 31.25 % attributing this impact to a great degree. We also explored respondents' willingness to pay (WTP) for a singular excursion to the grasslands. Participants expressed varying levels of WTP for CES use. Residents aged 30–40 exhibited the highest mean WTP, with 47.57 % indicating a WTP range of 2000 to 4000 RMB, and 16.50 % expressing a willingness to pay beyond the threshold of

4000 RMB. Conversely, those in the 50–60 age bracket demonstrated the lowest average WTP. WTP also varied among educational cohorts, with undergraduates manifesting the highest average WTP and high school students the lowest.

3.2. Spatial pattern of GCES supply

Spatial distribution of GCES supply in IMAR discerned that ST-V commanded paramount importance, eclipsing SB-V and SEH-V, while GH-V ranked the lowest based on grassland cultural value index (GCVI) (Fig. 5). Areas with high GCVI predominantly clustered around specific POIs, such as diverse landscapes, historical sites, temples, nature reserves and other scenic spots. 42.3 % of the territory fell within high ST-V supply, while 54 % showed a medium supply. ST-V exhibited high supply across most areas, particularly in the central and western regions of Huhehaote, Eerduosi, Baotou, Wulanchabu, as well as the central and eastern regions of Chifeng, Tongliao, and Hulunbeier. SB-V and SEH-V exhibited analogous spatial patterns, characterized by higher values distributed along the road network or water system. Areas with high SB-V and SEH-V supply covered nearly 10 % of the territory. However, the majority, exceeding 80 %, falls within the median level for both value types, indicating the need for further investment to unlock the cultural potential in these areas. Conversely, GH-V displayed limited variation in GCVI across the study area, with a prevailing prevalence of medium values and a conspicuous dearth of high values in most regions.

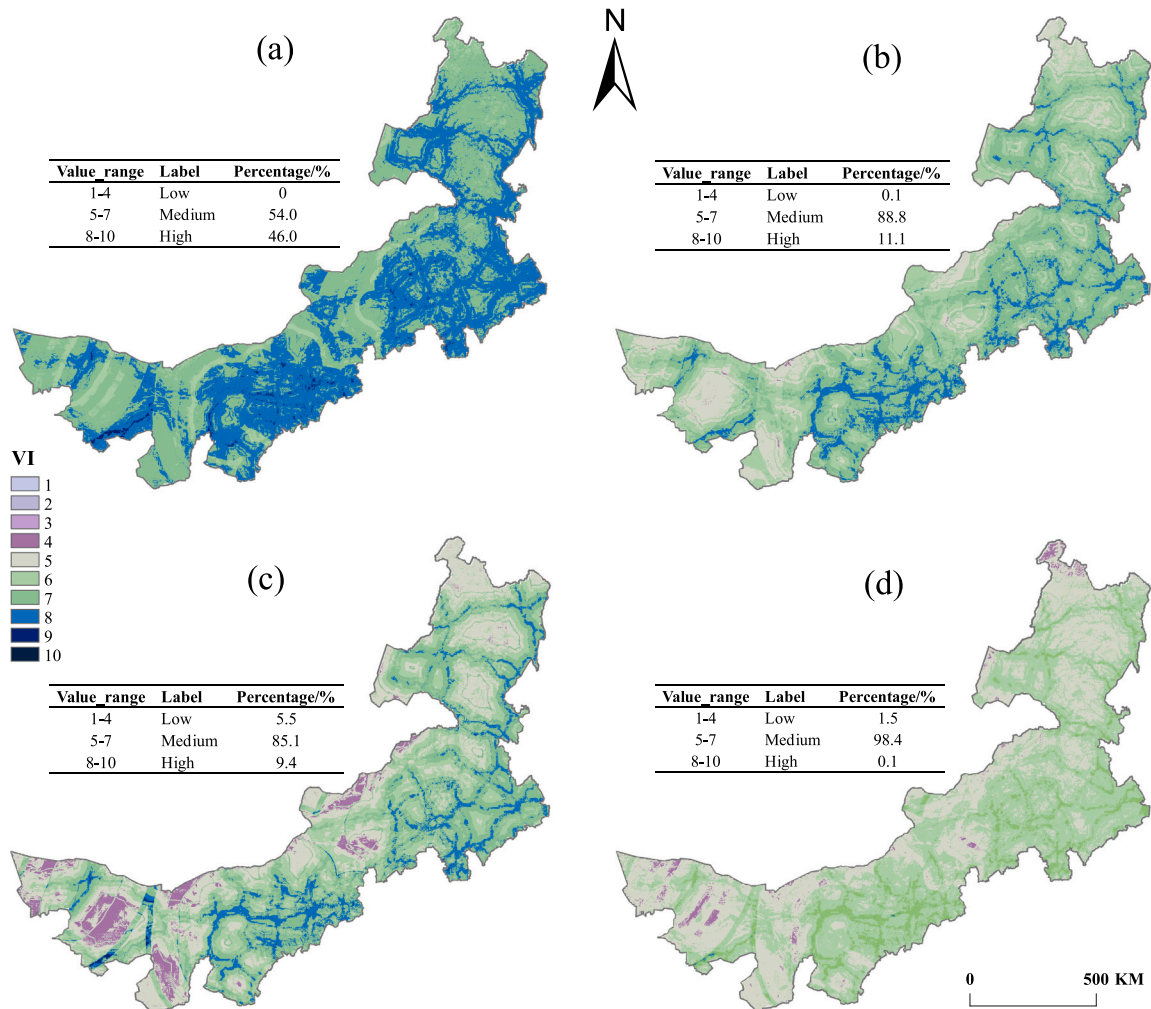


Fig. 5. Value index of grassland cultural ecosystem services (GCES) in the study area. (a) scenic travel value index, (b) spiritual belief value index, (c) science-education-history value index, (d) grassland recuperation value index.

Table 2
Importance and contribution rate /% of each environment variable.

Environment variables	ST-V		SB-V		SH-V		GH-V	
	Contribution	Importance	Contribution	Importance	Contribution	Importance	Contribution	Importance
DEM	6.6	11.1	5.8	9.6	2.4	5.6	6.9	10.4
DTG	0.4	0.02	1.0	0.8	0.9	0.9	0.6	0.9
DTR	57.3	52.7	62.2	52.0	64.2	57.3	54.3	48.4
DTW	19.2	16.7	18.0	16.6	18.0	17.6	18.3	16.9
LANDFORM	12.2	13.2	8.2	13.7	11.2	12.5	13.5	14.7
LULC	1.6	2.8	3.4	4.1	1.7	3.2	2.4	4.1
SLOPE	2.6	3.5	1.4	3.3	1.6	2.9	4.1	4.6

GCES were significantly impacted by natural environment and spatial distance, including DTR, DTW, LANDFORM, and DEM, collectively contributing to approximately 90 % of cultural value (Table 2). Notably, DTR exhibited the highest influence on the four cultural value, surpassing 50 % across all categories: ST-V (57.3 %), SB-V (62.2 %), SEH-V (64.2 %), and GH-V (54.3 %). DTW contributed approximately 20 % to these services, while LANDFORM and DEM accounted for roughly 10 % and 5 %, respectively. It was noticeable that GCES exhibited a negative correlation with increasing distance from roads,

and a downward trend then upward trend with increasing distance from water, highlighting the distinct role of the road and water network system in the grassland (Supplementary Fig. S2). Additionally, an altitude-dependent pattern was observed, with cultural value initially increasing, followed by a subsequent decrease with rising altitude, particularly high in the mid and low-altitude plains and hills. In summary, transportation conditions and associated travel costs were identified as the primary determinants of GCES value.

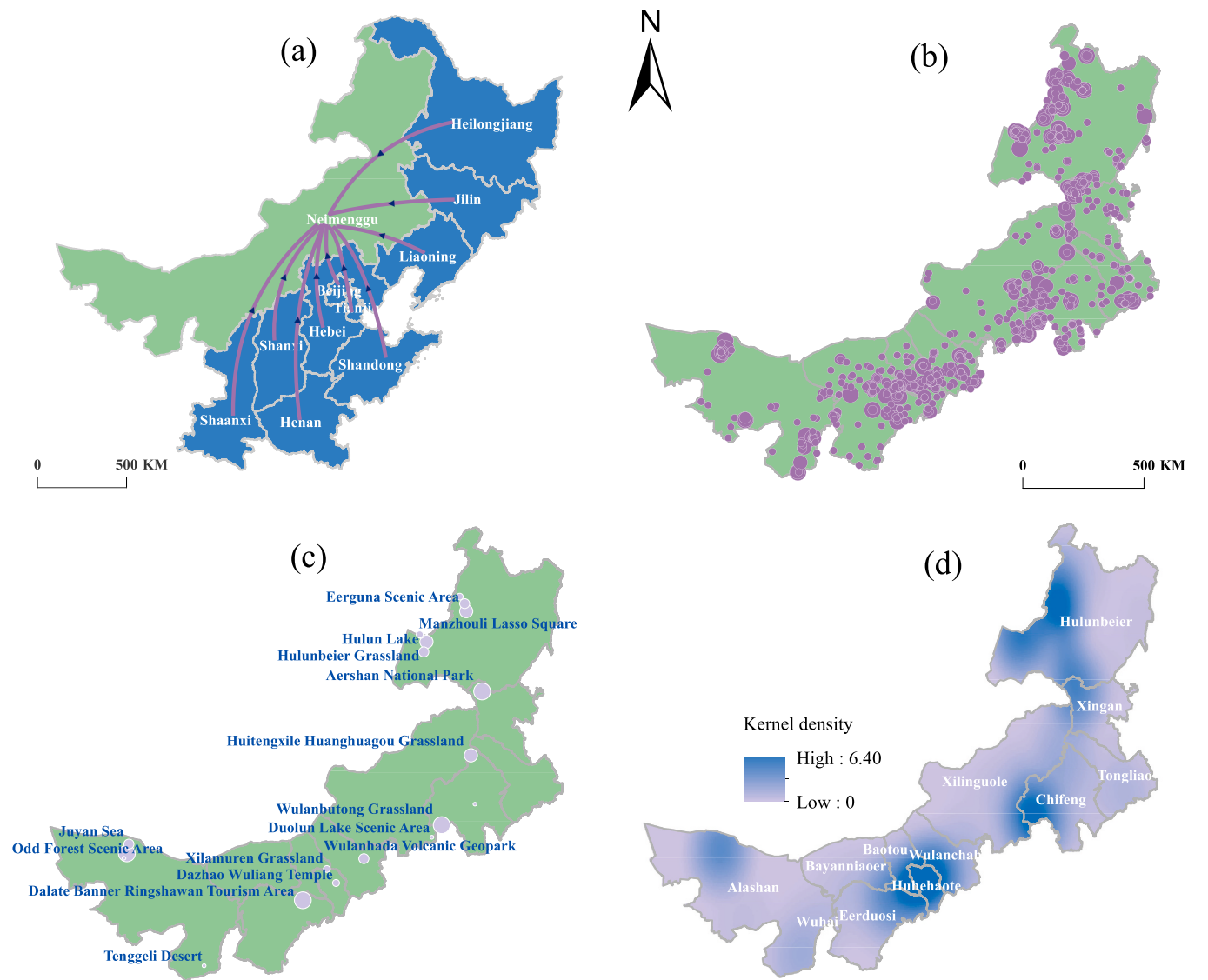


Fig. 6. Distribution of demand for grassland cultural ecosystem services (GCES). (a) The top ten distribution of domestic tourists, (b) Point of Interest (POI) of GCES demand, (c) GCES top 20 attractions distribution, (d) GCES demand kernel density result distribution.

3.3. Spatial pattern of GCES demand

Our analysis unveiled significant demand hotspots and illustrated the geographical spread of GCES (Fig. 6). The top ten domestic tourists arriving from outside IMAR mainly hailed from provinces neighboring IMAR, underscoring the appeal of GCES at adjacent spatial distances (Fig. 6a). Further investigation identified five key demanded CES-related destinations spanning from west to east: (1) Ejina Poplar Forest and Juyan Sea in Alashan League, (2) Dalate Banner Xiangshawan tourist area in Eerduosi City, Xilamuren Grassland in Baotou, and Dazhao Immense Temple in Huhehaote, (3) Wulanbutong Grassland in Chifeng and the scenic area of Duolun Lake in Xilinguole, (4) Huitengxile Huanghuagou Grassland tourist area in Xilinguole, (5) Hulunbeier Grassland and Hulun Lake in Hulunbeier, situated at the core of the radiation zone (Fig. 6c and Fig. 6d).

3.4. Patterns of GCES supply-demand match and mismatch

ESDR revealed a regional surplus of GCES supply over demand, with 65.99 % of areas demonstrating this excess (Fig. 7c). Within this surplus, 19.02 % exhibited overwhelming excess, signifying untapped potential. Conversely, 13.35 % exhibited demand surpassing supply, while only 20.66 % achieve a supply-demand equilibrium, underscoring significant imbalance. Huhehaote, with the highest demand, emerged with 75.51 % mismatch that demand exceeds supply (Fig. 7d), whereas Baotou (29.86 %), Hulunbeier (25.66 %), Wulanchabu (18.05 %), Chifeng (15.19 %), and Eerduosi (9.06 %) had relatively small proportions of supply shortfall. Tongliao and Wuhai stood out with excess supply, attributed to low demand due to accessibility or popularity. In other regions, a certain percentage exhibited a balanced supply and demand. Wulanchabu (30.62 %), Alashan (30.52 %), Baotou (23.22 %), and Hulunbeier (22.93 %) had >20 % of regions achieving parity, while other areas tended to have below 20 % balance.

The spatial distribution of GCES hot spots and cold spots in IMAR, as determined by Getis-Ord G_i^* analysis, is shown in Fig. 7e. Results showed that areas of excess supply (hot spots) and excess demand (cold spots) each constituted 26.03 % and 22 % at the 95 % significance level (p -value<0.05). Furthermore, regions exhibiting surplus supply are mainly dominated by highly natural landscapes centered around renowned attractions, nature parks, reserves, historical sites, and religious landmarks. Conversely, locales facing excess demand are situated in densely populated, economically developed urban centers or flat plains, exemplified by specific sections of cities such as Huhehaote, Hulunbeier, Baotou, Wulanchabu, or in geographical areas like Hulunbeier Grassland, Xilamuren Grassland, and Dalate Banner Xiangshawan tourist area. Spatially, a discernible overlap and dislocation emerge between the realms of surplus supply and excess demand.

4. Discussion

4.1. Pubic preferences for GCES

The overwhelming acknowledgment of various cultural aspects underscores the multifaceted role that grasslands play in the lives of individuals. Notably, scenic travel (91.18 %) emerged as the foremost cultural aspect (Fig. 4b), indicating a prevalent aesthetic and recreational appreciation for grasslands. This preference contrasts with lower inclinations toward science and education (39.34 %). Consequently, the perceived cultural value of grasslands predominantly links to their scenic and leisurely attributes rather than their educational or scientific dimensions. A substantial majority perceive grasslands as conducive to recuperation (98 %), spiritual belief (76.47 %), overall sense of well-being (99.63 %). This underscores the perceived positive impact of grasslands' potential for serving as holistic environments that contribute to physical and mental well-being, aligning seamlessly with existing literature on the therapeutic and restorative benefits of nature-based

experiences (Triguero-Mas et al., 2015).

Upon further analysis, substantial disparities emerged in the selection of and readiness to pay for cultural services among varying age cohorts, educational backgrounds, and other socio-demographic traits. This underscores the potential influence of individual background factors on the demand and valuation of these services (Pinto et al., 2023). Our investigation into respondents' WTP unveiled intriguing trends. Individuals aged 30–40 demonstrated the highest mean WTP (Fig. 4c), signaling a considerable financial commitment to grassland experiences. This discernment alludes to the confluence of economic capacity and cultural significance intertwined with these experiences within the specified age demographic. Moreover, a conspicuous discrepancy in WTP emerges among educational groups, with undergraduates displaying the utmost average WTP and high school students the lowest (Fig. 4d). This incongruity may emanate from variances in disposable income, perceived value, or the contextual pertinence of grassland experiences within these disparate educational milieus. In light of the survey outcomes, we propose that pertinent organizations should prioritize understanding respondents' preferences and requirements while promoting cultural services related to grasslands. Moreover, the enhancement of public cognizance and participation in these services could be actualized through the implementation of diverse pricing strategies or the facilitation of cultural and educational events.

4.2. Spatial pattern and influencing factors of GCES

Quantitative analysis of GCES values at the grid scale reveals a nuanced distribution pattern, emphasizing the paramount importance of ST-V over SB-V and SEH-V, with GH-V registering the lowest based on GCVI. ST-V maintains its preeminence, aligning seamlessly with qualitative interview results we discussed earlier, highlighting the prominence of ecotourism in actualizing the value of grassland culture. GH-V, a recently proposed value type, currently lacks widespread recognition and requires ongoing exploration to unlock its potential in the future. The variations in values across different cultural services stem from objective disparities in the inherent capacities of grassland ecosystems to provide diverse cultural services and subjectively from differences in public perceptions and preference choices (Pinto et al., 2023).

We observed that certain cities in the central-western and central-eastern regions host a high supply of GCES, converging around specific POIs that span diverse landscapes, historical sites, temples and nature reserves. Areas with high demand are situated in densely populated urban hubs or flat plains, consistent with earlier research (Schirpke et al., 2018). Notably, cities like Huhehaote, Eerduosi, Baotou, and Wulanchabu in the central-west, boasting robust GDP in IMAR, exhibit abundant tourism and cultural resources alongside well-developed public infrastructure, fostering the delivery of exemplary cultural services. Our findings align with prior research by demonstrating that regions endowed with abundant natural landscapes and attractions exhibit elevated CES (Schirpke et al., 2018; Lorilla et al., 2019). Our observations also reveal parallel patterns in SB-V and SEH-V, characterized by heightened values aligning with the road network or water systems. This phenomenon is attributed to the role of road and waterway networks in enhancing ecosystem connectivity, thereby influencing human activities and consequently impacting the provisioning of ecosystem services.

Moreover, Our analysis has pinpointed DTR as the most influential factor contributing to GCES, in line with previous studies emphasizing the pivotal role of road accessibility in CES (Ala-Hulkko et al., 2016; Van Zanten et al., 2016; Kulczyk et al., 2018; Shan et al., 2019). In our investigation, GCES supply was found to decrease with increasing distance from roads and to decrease then rise with increasing distance from water, a trend that may differ from other CES studies. Shan et al. (2019) observed that recreational supply of farmlands exhibited an initial increase and subsequent decrease with increasing distance to city centers, villages and water. Zuo and Zhang (2023) discovered that most CES exhibit a downward trend with increasing distance from national

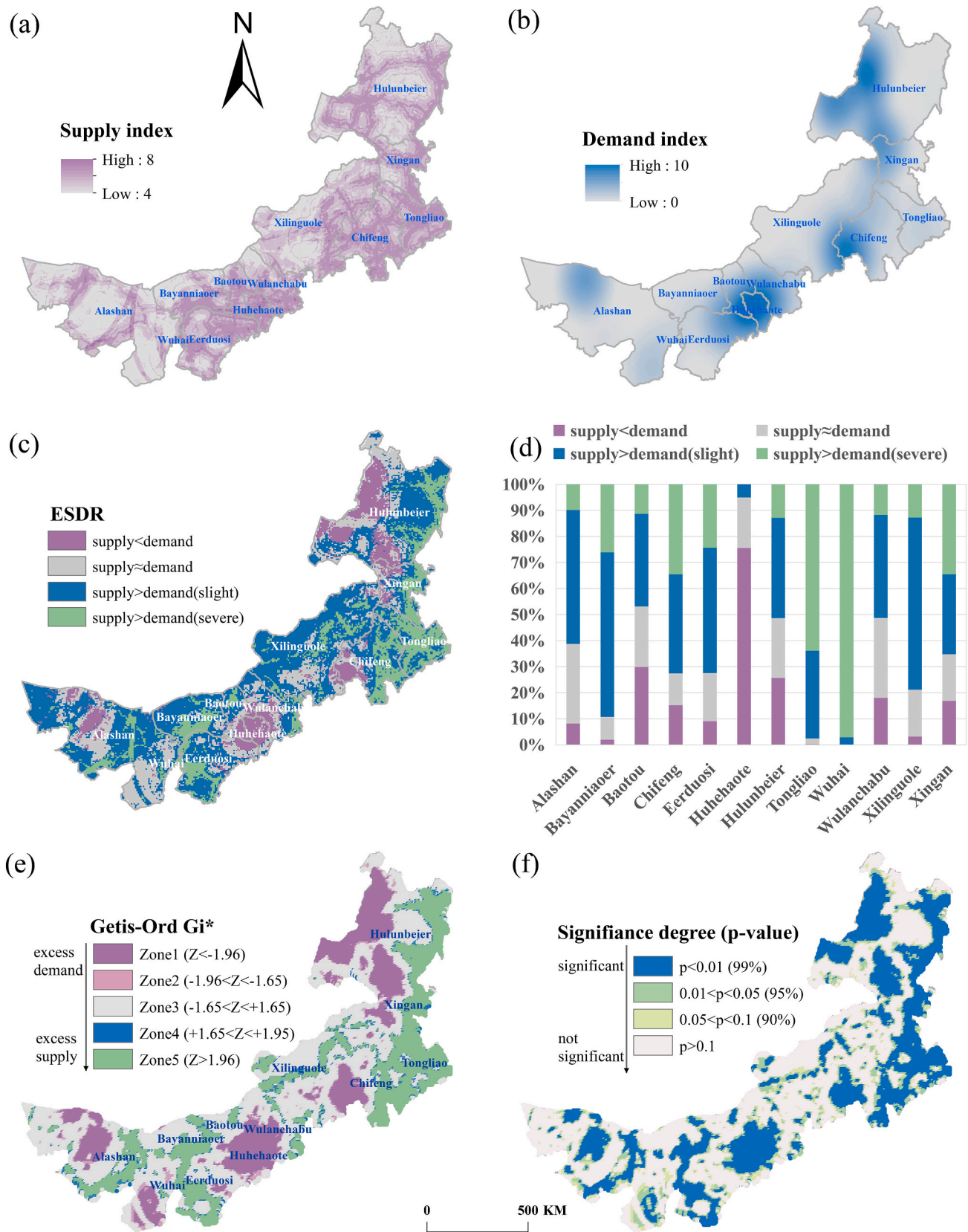


Fig. 7. Matching results of grassland cultural ecosystem services (GCES) supply-demand. (a) GCES supply value index, (b) GCES demand index, (c) GCES supply-demand matching pattern based on ESDR, (d) GCES statistical supply-demand matching by region, (e) Distribution of GCES hot spot and cold spot based on z-value of Getis-Ord G_i^* , (f) Significance degree (p-value) of Getis-Ord G_i^* .

roadways, spiritual and historical culture decline with an increase in DTW, while aesthetic and recreation demonstrate an initial ascent followed by a subsequent decline with increasing distance from township centers. Contrastingly, [Pinto et al. \(2023\)](#) reported that the appeal of cultural services to individuals positively correlates with distance within a specific range, with greater distances associated with increased relevance. Additionally, they highlighted that factors other than distance play a more significant role in influencing CES, with areas of high CES concentration in more urbanized regions. In summary, the impact mechanism of CES is intricate and varies due to disparities in the geographic environment, cultural context, and other factors among the study subjects.

4.3. Mismatches of GCES supply-demand

Our findings reveal a substantial surplus of supply exceeding demand for GCES in the majority of IMAR, aligning with prior studies emphasizing the pivotal role of grasslands in providing diverse regulatory and cultural services ([Shen et al., 2023](#)). This surplus arises from several factors: First, grasslands encompass diverse natural landscapes, cultural legacies, and traditional folk cultures, constituting a rich potential for supplying cultural services. These landscapes encompass national grassland nature parks, nature reserves, scenic areas, historic sites, and religious venues, including six 5 A and twenty-eight 4 A scenic spots. Moreover, intangible cultural heritage, such as festivals, customs, athletics, grassland handicrafts, literature and dance, contributes significantly to this cultural value. Second, limited popularity of certain highly natural and remote areas has diminished tourist demand. Enhanced cultural promotion efforts are therefore imperative in the future. Third, accessibility emerges as a key factor influencing public demand for CES ([Syrbe and Grunewald, 2017](#)). Due to the sparsity of inhabitants in IMAR grassland areas, inadequate transportation infrastructure and underdeveloped public facilities naturally translate to diminished tourist demand. The amalgamation of these factors culminates in a scenario characterized by a surplus of supply exceeding demand.

Spatially, GCES supply and demand in IMAR show pronounced spatial misalignment, with only 20.66 % achieving equilibrium. This misalignment is essentially a mismatch between the resource endowments of the supplying regions and the actual conditions of access to those resources ([Syrbe and Walz, 2012](#)). Only when CES reach the demand area through the flow of space can their service value be realized ([Syrbe and Grunewald, 2017](#)). However, the spatial flow from supply to demand areas for GCES in IMAR encounters impediments, preventing the precise and dynamic spatial transfer of CES from the provisioning region to the beneficiary region. This hinders the effective conversion rate from supply to demand ([Vallecillo et al., 2019](#)). In summary, the spatial incongruity between CES supply and demand is influenced by multiple factors, including infrastructure availability, transportation accessibility, proximity of cultural value points to roads, city centers, and water bodies, as well as the level of urbanization. Appropriate policies and interventions are necessary to rectify this mismatch ([Pinto et al., 2023](#)). Furthermore, future research endeavors must delve into the intricate causal feedback mechanisms and flows between the supply and demand of CES within coupled human-nature systems ([Syrbe and Walz, 2012](#)).

4.4. Uncertainties and limitations

Our analysis faces limitations due to data availability and current quantification methods, potentially affecting result reliability. First, the SolVES model, reliant on survey data for GCES quantification, is susceptible to bias deriving from (1) respondent subjectivity, their ability to effectively comprehend the questionnaire's content, and demographic factors; (2) the scientific rigor of questionnaire design; (3) the spatial distribution homogeneity of the sample, and the relevance of the survey implementation. This study lacked focused survey objectives and

differentiation between various groups. For example, distinguishing temple devotees, religious groups, and herders is pivotal for a comprehensive grasp of cultural values and spirituality quantification. Nevertheless, due to objective constraints, an online platform was utilized for surveys, challenging control over participant characteristics.

Furthermore, the SolVES model, typically applied at small scales, such as forest parks, urban parks, and rural areas, has demonstrated efficacy. This study, however, applied SolVES model to an entire administrative unit. The training and test AUC values for the four categories of GCES were all in the range of 0.5–0.7, representing low accuracy. Actually, we have tried to make some optimization adjustments during the investigation phase. Although the survey group's size remained consistent with previous studies, a refinement in questionnaire design replaced individual value point annotations with the selection of value POI. This alteration facilitated the vectorization of value points, resulting in a notable increase in their quantity and mitigating the limitations of model applicability as research scales expand. Subsequent inquiries should seek enhancements to improve SolVES model's suitability for larger scales and its effectiveness in quantifying CES.

Finally, the reliability, accuracy, and sufficiency of social media data can significantly impact GCES demand quantification. In this study, horizontal life review websites and vertical tourism professional review websites were selected as representative sources. However, since these websites are predominantly used by a younger demographic, they may not fully capture the demand from certain groups that benefit from GCES. This could lead to an underestimation of the actual demand for GCES. Meanwhile, our study used the size of the tourist population to estimate GCES demand, potentially introducing deviation in our estimation of actual GCES demand. A broad spectrum of socio-cultural and economic factors including gender, age, income, culture, and education can have an impact on the demand. Therefore, future research must expand the sources of quantitative demand data and develop appropriate methods with considering more factors into demand function to minimize the uncertainty of demand evaluation.

4.5. Implication for planning and management

Tailored management policies are vital to address the GCES supply-demand imbalance, crucial for achieving spatial objectives in social-ecological system sustainability. Protecting areas with higher GCVI to preserve cultural value hot spots is essential, alongside efforts to cultivate cultural brands supporting high supply capacities. Increasing investment in lower GCVI areas and formulating development policies can enhance their supply capacities. Moreover, areas with GCES oversupply should focus on enhancing road accessibility, while those facing excess demand should prioritize resource allocation to prevent surpassing environmental carrying capacity.

Specifically: (1) Regions with limited supply should prioritize building grassland tourism infrastructure, such as constructing cultural amenities, attracting investments in cultural assets, improving local transportation, and strengthening the protection of environmentally fragile areas while mitigating ecological impact. (2) Develop popular cultural products, encompassing cultural sightseeing tourism and recreational activities like sports, festivals, music, dance and art performances. (3) Cultivate specialized cultural products. Firstly, center efforts on religious tourism, capitalizing on the profound influence of Shamanism and Tibetan Buddhism on the lifestyle and mindset of herdsmen. This can be manifested at locations like Da Zhao, Xiao Zhao, pagodas, and temples, where religious artifacts, structures, ceremonies, and influences can be fashioned into unique tourism products. Secondly, explore grassland recuperation tourism in areas with suitable natural environments, climate conditions, and rich cultural significance. Develop personalized experiences that allow visitors to unwind, alleviate stress, and immerse themselves in the authentic grassland lifestyle. Thirdly, engage in science-history research tourism. In some typical grassland regions, specialized scientific research tourism can be

developed, such as investigating grassy meadows, typical grasslands, and degraded grasslands. In some historical relic sites and locales with significant human events, historical research tourism can be developed, such as Dayao cultural site, Ashan ruins, Yinshan Rock Paintings, etc. (4) Implement regional product differentiation. To create unique regional offerings, adopt a spatial differentiation matching approach rooted in supply-demand analysis. In the western, central, and eastern regions with varying natural landscapes, develop distinctive regional products to avoid low-level duplication and achieve genuine cultural improvement in tourist offerings.

5. Conclusion

This investigation employs innovative technical methodologies for the quantitative assessment of GCES to delineate areas characterized by high supply and demand radiation, as well as those exhibiting surplus, equilibrium, or deficiency. Noteworthy cultural value sites congregate around prominent attractions, nature reserves, historical landmarks, and religious sites. Our scrutiny underscores the pivotal role of road accessibility as a determining factor influencing GCES levels. Furthermore, our findings reveal an overarching surplus in GCES supply relative to prevailing demand, indicative of significant untapped potential within the realm of GCES. To optimize the efficient utilization of GCES, we propose judicious resource allocation, the augmentation of public infrastructure to fortify ecological carrying capacity in high-demand regions, improvement of transportation conditions in areas marked by high supply, and the implementation of sustainable tourism strategies for grassland development. In conclusion, our study contributes innovative perspectives to CES research, furnishing a comprehensive framework for comprehending CES and offering insightful guidance to policymakers in shaping sustainable development initiatives. Nevertheless, we acknowledge the imperative for future endeavors to address data constraints and refine quantitative analytical methodologies for CES.

CRedit authorship contribution statement

Lan Wang: Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **Lin Huang:** Conceptualization, Methodology, Supervision, Validation, Writing – review & editing. **Wei Cao:** Validation, Writing – review & editing. **Jun Zhai:** Supervision, Writing – review & editing. **Jiangwen Fan:** Resources, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This research was supported by the National Key Research and Development Program of the 14th Five-Year Plan – "Technical Methods and Value Realization Mechanisms of Accounting for Regional Ecological Products" (grant number: 2022YFF1301800). We would like to thank the two anonymous reviewers for their important and insightful comments and suggestions which helped improve the manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2023.169255>.

References

- Ala-Hulkko, T., Kotavaara, O., Alahuhta, J., Helle, P., Hjort, J., 2016. Introducing accessibility analysis in mapping cultural ecosystem services. *Ecol. Indic.* 66, 416–427.
- Bagstad, K.J., Semmens, D.J., Ancona, Z.H., et al., 2017. Evaluating alternative methods for biophysical and cultural ecosystem services hotspot mapping in natural resource planning. *Landsc. Ecol.* 32 (1), 77–97. <https://doi.org/10.1007/s10980-016-0430-6>.
- Burkhard, B., Kroll, F., Nedkov, S., et al., 2012. Mapping ecosystem service supply, demand and budgets. *Ecol. Indic.* 21, 17–29. <https://doi.org/10.1016/j.ecolind.2011.06.019>.
- Chen, J., Jiang, B., Bai, Y., Xu, X., Alatalo, J.M., 2019. Quantifying ecosystem services supply and demand shortfalls and mismatches for management optimisation. *Sci. Total Environ.* 650 (1), 1426–1439. <https://doi.org/10.1016/j.scitotenv.2018.09.126>.
- Costanza, R., d'Arge, R., de Groot, R., Farber, S., Grasso, M., Hannon, B., Limburg, K., Naem, S., O'Neill, R.V., Paruelo, J., Raskin, R.G., Sutton, P., van den Belt, M., 1997. The value of the world's ecosystem services and natural capital. *Nature* 387 (1), 253–260.
- Crouzat, E., De Frutos, A., Grescho, V., Carver, S., Büermann, A., Carvalho-Santos, C., Bonn, A., 2022. Potential supply and actual use of cultural ecosystem services in mountain protected areas and their surroundings. *Ecosyst. Serv.* 53, 101395.
- Daily, G., 1997. *Nature's Services: Societal Dependence on Natural Ecosystems*. Island Press, Washington DC.
- Daniel, T.C., Muhar, A., Arnberger, A., et al., 2012. Contributions of cultural services to the ecosystem services agenda[J]. *Proc. Natl. Acad. Sci. U. S. A.* 109 (23), 8812–8819. <https://doi.org/10.1073/pnas.1114773109>.
- Das, A., Das, M., Saha, S., et al., 2023. Impact of COVID-19 pandemic on cultural ecosystem services from urban green spaces: a case from English Bazar Urban Agglomeration. *Eastern India. Environ. Sci. Pollut. Res.* <https://doi.org/10.1007/s11356-023-26919-0>.
- De Groot, R.S., Wilson, M.A., Boumans, R.M.J., 2002. A typology for the classification, description and valuation of ecosystem functions, goods and services. *Ecol. Econ.* 41 (3), 393–408.
- De Valck, J., Broekx, S., Liekens, I., et al., 2016. Contrasting collective preferences for outdoor recreation and substitutability of nature areas using hot spot mapping. *Landsc. Urban Plan.* 151, 64–78.
- Díaz, S., Settele, J., Brondizio, E.S., et al., 2019. Pervasive human-driven decline of life on earth points to the need for transformative change. *Science* 366 (6471), eaax3100.
- Fletcher, R., Baulcomb, C., Hall, C., Hussain, S., 2014. Revealing marine cultural ecosystem services in the Black Sea. *Mar. Policy* 50, 151–161.
- Fsa, B., Jsc, D., Jcba, B., 2021. Assessment and prioritization of cultural ecosystem services in the Sahara-Sahelian region[J]. *Sci. Total Environ.* <https://doi.org/10.1016/j.scitotenv.2021.146053>.
- Getis, A., Ord, J.K., 1992. The analysis of spatial association by use of distance statistics. *Geogr. Anal.* 24, 3. <https://doi.org/10.1111/j.1538-4632.1992.tb00261.x>.
- Gosal, A.S., Geijzendorffer, I.R., V'aclovík, T., et al., 2019. Using social media, machine learning and natural language processing to map multiple recreational beneficiaries. *Ecosyst. Serv.* 38 (June 2018), 100958. <https://doi.org/10.1016/j.ecoser.2019.100958>.
- Gould, K.R., Lincoln, K.N., 2017. Expanding the suite of cultural ecosystem services to include ingenuity, perspective, and life teaching. *Ecosyst. Serv.* 25, 117–127. <https://doi.org/10.1016/j.ecoser.2017.04.002>.
- Hernandez-Morcillo, M., Plieninger, T., Bieling, C., 2013. An empirical review of cultural ecosystem service indicators. *Ecol. Indic.* 29, 434–444. <https://doi.org/10.1016/j.ecolind.2013.01.013>.
- Jiang, C., et al., 2021. Identifying the spatial disparities and determinants of ecosystem service balance and their implications on land use optimization. *Sci. Total Environ.* 793, 148472.
- Karasov, O., Heremans, S., Külvik, M., Domnich, A., Burdun, I., Kull, A., Uuema, E., 2022. Beyond land cover: how integrated remote sensing and social media data analysis facilitates assessment of cultural ecosystem services. *Ecosyst. Serv.* 53, 101391.
- Kosanic, A., Petzold, J., 2020. A systematic review of cultural ecosystem services and human wellbeing. *Ecosyst. Serv.* 45, 101168. <https://doi.org/10.1016/j.ecoser.2020.101168>.
- Kulczyk, S., Woźniak, E., Derek, M., 2018. Landscape, facilities and visitors: an integrated model of recreational ecosystem services. *Ecosyst. Serv.* 31 (Part C), 491–501. <https://doi.org/10.1016/j.ecoser.2018.02.016>.
- Li, J., Fang, J., 2017. *A Preface to the Research on the Strategy of Grassland Ecology and Food Safety in China*, vol. 3. Science Press, Beijing.
- Li, J., Jiang, H., Bai, Y., Alatalo, J.M., Li, X., Jiang, H., Liu, G., Xu, J., 2016. Indicators for spatial-temporal comparisons of ecosystem service status between regions: a case study of the Taihu River Basin. *China. Ecol. Indic.* 60, 1008–1016. <https://doi.org/10.1016/j.ecolind.2015.09.002>.
- Liao, J., Liao, Q., Wang, W., et al., 2023. Quantifying and mapping landscape value using online texts: a deep learning approach. *Appl. Geogr.* 154. <https://doi.org/10.1016/j.apgeog.2023.102950>.
- Lingua, F., Coops, N.C., Griess, V.C., 2023. Assessing forest recreational potential from social media data and remote sensing technologies data. *Ecol. Indic.* 149. <https://doi.org/10.1016/j.ecolind.2023.110165>.

- Liu, H., Remme, R.P., Hamel, P., Nong, H., Ren, H., 2020. Supply and demand assessment of urban recreation service and its implication for greenspace planning—a case study on Guangzhou. *Landsc. Urban Plan.* 203 (2020), 103898.
- Liu, Zhong-ling, 2010. Ecological connotation of Mongolian grassland traditional culture [J]. *Pratacult. Sci.* 27 (01), 1–3.
- Lorilla, R.S., Kalogirou, S., Poirazidis, K., et al., 2019. Identifying spatial mismatches between the supply and demand of ecosystem services to achieve a sustainable management regime in the Ionian Islands (Western Greece). *Land Use Policy* 88, 104171. <https://doi.org/10.1016/j.landusepol.2019.104171>.
- MEA (Millennium Ecosystem Assessment), 2005. *Ecosystems and Human Well-Being: Biodiversity Synthesis*. Island Press, Washington DC.
- Mitra, Ghasemi, Zabih, Charrayh, González-García, Alberto, 2023. Mapping cultural ecosystem services provision: An integrated model of recreation and ecotourism opportunities. In: *Land Use Policy*, Volume 131, p. 106732. ISSN 0264-8377. <https://doi.org/10.1016/j.landusepol.2023.106732>.
- Niu, J.M., 2000. Relationship between main vegetation types and climatic factors in Inner Mongolia [J]. *Chin. J. Appl. Ecol.* 11 (1), 47–52.
- Pan, Z., He, J., Liu, D., Wang, J., Guo, X., 2021. Ecosystem health assessment based on ecological integrity and ecosystem services demand in the middle reaches of the Yangtze river economic belt. *China. Sci. Total Environ.* 774, 144837.
- Paracchini, M.L., Zulian, G., Kopperoinen, L., et al., 2014. Mapping cultural ecosystem services: A framework to assess the potential for outdoor recreation across the EU. *Ecol. Indic.* 45, 371–385.
- Phillips, S.J., Dudík, M., 2008. Modeling of species distributions with Maxent—new extensions and a comprehensive evaluation. *Ecography* 31 (2), 161–175. <https://doi.org/10.1111/j.0906-7590.2008.5203.x>.
- Pinto, L.V., Inacio, M., Bogdzewic, K., Kalinauskas, M., Gomes, E., Pereira, P., 2023. Factors affecting cultural ecosystem services use in Vilnius (Lithuania): a participatory mapping survey approach. *HELIYON* 9 (4). <https://doi.org/10.1016/j.heliyon.2023.e15384>.
- Plieninger, T., Bieling, C., Fagerholm, N., et al., 2015. The role of cultural ecosystem services in landscape management and planning. *Curr. Opin. Environ. Sustain.* 14, 28–33. <https://doi.org/10.1016/j.cosust.2015.02.006>.
- Qiqi Zhao, Yan Chen, Keshava Pallavi Gone, et al. Modelling cultural ecosystem services in agricultural dykelands and tidal wetlands to inform coastal infrastructure decisions: A social media data approach. *Marine Policy*, Volume 150, 2023, 105533, ISSN 0308-597X, doi:<https://doi.org/10.1016/j.marpol.2023.105533>.
- Satz, D., Gould, R.K., Chan, K.M.A., et al., 2013. The challenges of incorporating cultural ecosystem services into environmental assessment. *Ambio* 42 (6), 675–684. <https://doi.org/10.1007/s13280-013-0386-6>.
- Schirpke, U., Meisch, C., Marsoner, T., et al., 2018. Revealing spatial and temporal patterns of outdoor recreation in the European Alps and their surroundings. *Ecosyst. Serv.* 31 (C), 336–350. <https://doi.org/10.1016/j.ecoser.2017.11.017>.
- Schrter, M., Barton, D.N., Remme, R.P., et al., 2014. Accounting for capacity and flow of ecosystem services: a conceptual model and a case study for Telemark, Norway [J]. *Ecol. Indic.* 36, 539–551.
- Shan, He, Yue, Su, Shahtahmassebi, Amir Reza, et al., 2019. Assessing and mapping cultural ecosystem services supply, demand and flow of farmlands in the Hangzhou metropolitan area, China. *Sci. Total Environ.* 692, 756–768. <https://doi.org/10.1016/j.scitotenv.2019.07.160>.
- Shaw, E., Kumar, V., Lange, E., et al., 2016. Exploring the utility of Bayesian networks for modelling cultural ecosystem services — a canoeing case study. *Sci. Total Environ.* 540, 71–78. <https://doi.org/10.1016/j.scitotenv.2015.08.027>.
- Shen, J., Li, S., Wang, H., et al., 2023. Understanding the spatial relationships and drivers of ecosystem service supply-demand mismatches towards spatially-targeted management of social-ecological system [J]. *J. Clean. Prod.* 406, 136882 <https://doi.org/10.1016/j.jclepro.2023.136882>.
- Sherrouse, B.C., Clement, J.M., Semmens, D.J., 2011. A GIS application for assessing, mapping, and quantifying the social values of ecosystem services [J]. *Appl. Geogr.* 31, 748–760. <https://doi.org/10.1016/j.apgeog.2010.08.002>.
- Shiting, Meng, Qingxu, Huang, Ling, Zhang, et al., 2020. Matches and mismatches between the supply of and demand for cultural ecosystem services in rapidly urbanizing watersheds: A case study in the Guanting Reservoir basin, China - ScienceDirect [J]. *Ecosyst. Serv.* 45 <https://doi.org/10.1016/j.ecoser.2020.101156> [2023-09-16].
- Sitong, Huang, Tian, Tian, Lingge, Zhai, et al., 2023. Understanding the dynamic changes in wetland cultural ecosystem services: Integrating annual social media data into the SolVES, *Applied Geography*, Volume 156. <https://doi.org/10.1016/j.apgeog.2023.102992>, 102992, ISSN 0143-6228.
- Small, N., Munday, M., Durance, I., 2017. The challenge of valuing ecosystem services that have no material benefits. *Glob. Environ. Chang.* 44, 57–67. <https://doi.org/10.1016/j.gloenvcha.2017.03.005>.
- Swets, J.A., 1988. Measuring the accuracy of diagnostic systems. *Science* 240 (4857), 1285–1293. <https://doi.org/10.1126/science.3287615>.
- Syrbe, R.-U., Grunewald, K., 2017. Ecosystem service supply and demand – the challenge to balance spatial mismatches. *Int. J. Biodivers. Sci. Ecosyst. Serv. Manag.* 13 (2), 148–161. <https://doi.org/10.1080/21513732.2017.1407362>.
- Syrbe, R.-U., Walz, U., 2012. Spatial indicators for the assessment of ecosystem services: providing, benefiting and connecting areas and landscape metrics. *Ecol. Indic.* 21, 80–88. <https://doi.org/10.1016/j.ecolind.2012.02.013>.
- Tenerelli, P., Demsar, U., Luque, S., 2016. Crowd sourcing indicators for cultural ecosystem services — a geographically weighted approach for mountain landscapes. *Ecol. Indic.* 64, 237–248. <https://doi.org/10.1016/j.ecolind.2015.12.042>.
- Teoh, S.H.S., Symes, W.S., Sun, H., et al., 2019. A global meta-analysis of the economic values of provisioning and cultural ecosystem services. *Sci. Total Environ.* 649, 1293–1298. <https://doi.org/10.1016/j.scitotenv.2018.08.422>.
- Toivonen, T., Heikinheimo, V., Fink, C., et al., 2019. Social media data for conservation science: a methodological overview. *Biol. Conserv.* 233 (January), 298–315. <https://doi.org/10.1016/j.biocon.2019.01.023>.
- Triguero-Mas, M., Dadvand, P., Cirach, M., Martínez, D., Medina, A., Mompert, A., Basagana, X., Grazuleviciene, R., Nieuwenhuijsen, M.J., 2015. Natural outdoor environments and mental and physical health: relationships and mechanisms. *Environ. Int.* 77, 35–41.
- Vallecillo, S., La Notte, A., Zulian, G., et al., 2019. Ecosystem services accounts: valuing the actual flow of nature-based recreation from ecosystems to people. *Ecol. Model.* 392, 196–211. <https://doi.org/10.1016/j.ecolmodel.2018.09.023>.
- Van Zanten, B.T., Van Berkel, D.B., Meentemeyer, R.K., et al., 2016. Continental-scale quantification of landscape values using social media data. *Proc. Natl. Acad. Sci. U. S. A.* 113, 12974–12979.
- Wei, H., Fan, W., Wang, X., et al., 2017. Integrating supply and social demand in ecosystem services assessment: a review [J]. *Ecosyst. Serv.* 25, 15–27. <https://doi.org/10.1016/j.ecoser.2017.03.017>.
- Xin, Nie, Wang, Qiaoling, Wu, Jianxian, et al., 2022. Effectively enhancing perceptions of cultural ecosystem services: A case study of a karst cultural ecosystem. *J. Environ. Manag.* 315 <https://doi.org/10.1016/j.jenvman.2022.115189>, 115189, ISSN 0301-4797.
- Yiyao, Wang, Kiichiro, Hayashi, 2023. Methodological development of cultural ecosystem services evaluation using location data. *J. Clean. Prod.* 396 <https://doi.org/10.1016/j.jclepro.2023.136523>, 136523, ISSN 0959-6526.
- Yoshimura, N., Hiura, T., 2017. Demand and supply of cultural ecosystem services: use of geotagged photos to map the aesthetic value of landscapes in Hokkaido. *Ecosyst. Serv.* 24, 68–78. <https://doi.org/10.1016/j.ecoser.2017.02.009>.
- Zuo, Y., Zhang, L., 2023. Research on local ecosystem cultural Services in the Jiangnan Water Network Rural Areas: a case study of the ecological green integration demonstration zone in the Yangtze River Delta. *China. Land* 12, 1373. <https://doi.org/10.3390/land12071373>.